# Beta Distribution

There are two types of Beta distribution i). Beta distribution of First kind and ii), beta distribution of second kind.

## Beta Distribution of First Kind

A continuous r.v. X is said to follow Beta distribution of first kind if the pdf of X is as follows

$$f(x)=\begin{cases}\dfrac{1}{B(\mu,\nu)}x^{\mu-1}(1-x)^{\nu-1};(\mu,\nu)>0;0<\mathrm{x}<1\\ 0;\text{otherwise}\end{cases}$$

Here $B(\mu,\nu)$ is Beta function. It is a two parameters distribution $\mu,\nu$ and denoted as $X\sim\beta_1(\mu,\nu)$.

### Mean and Variance

mean = $\mu=\dfrac{\mu}{\mu+\nu}$ and variance is $\sigma^2=\dfrac{\mu\nu}{(\mu+\nu)^2(\mu+\nu+1)}$

## Beta Distribution of Second Kind

A continuous r.v. X is said to follow Beta distribution of second kind if the pdf of X is as follows

$$f(x)=\begin{cases}\dfrac{1}{B(\mu,\nu)}\dfrac{x^{\mu-1}}{(1+x)^{\mu+\nu}};(\mu,\nu)>0;0<\mathrm{x}<1\\ 0;\text{otherwise}\end{cases}$$

Here $B(\mu,\nu)$ is Beta function. It is a two parameters distribution $\mu,\nu$ and denoted as $X\sim\beta_2(\mu,\nu)$.

beta distribution of second kind can be transformed to Beta distribution of first kind using the transformation $1+x=\dfrac{1}{y}\Rightarrow y=\dfrac{1}{1+x}$

### Mean and Variance

mean = $\mu=\dfrac{\mu}{\nu-1}$ and variance is $\sigma^2=\dfrac{\mu(\mu+\nu-1)}{(\nu-1)^2(\nu-2)}$